

Generating Shakespearean Text with a Character-Level Transformer

Generative AI Analysis and Ethics Report

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1. Overview

This project builds a small generative text system. The task is to write new dramatic dialogue in the style of Shakespeare, one character at a time, after learning from the Tiny Shakespeare corpus of about 1.1 million characters. The model is a decoder-only Transformer (Vaswani et al., 2017) implemented from scratch in PyTorch (Paszke et al., 2019). I chose a Transformer for text because attention lets the model relate any character in its context to any earlier one, the design that underlies current text generators (Radford et al., 2019). The report covers the data, the model and its training, an evaluation of the generated text, and an assessment of ethical risks that arise from this particular model and corpus.

2. Dataset or Prompt Description

The data are the Tiny Shakespeare corpus, a single plain-text file of 1,115,394 characters drawn from Shakespeare's plays and distributed with the public *char-rnn* repository (Karpathy, 2015). The plays are in the public domain, and the repository that distributes the file carries an MIT licence. I could not verify from the repository which edition of the plays the file reproduces, which is a documentation gap I note under limitations. The corpus uses 65 distinct characters, contains about 7,960 speaker labels (a name in capitals followed by a colon) and is written in Early Modern English. No prompts were used beyond the opening text supplied at generation time. The notebook downloads the file and verifies its SHA-256 checksum, so every rerun uses identical text.

Preparation. Each distinct character became a token, so the model needs no tokeniser and cannot meet an unknown symbol. Case was preserved because capitals mark speaker names. The first 90 per cent of the text (1,003,854 characters) was used for training and the final 10 per cent (111,540 characters) for validation, a contiguous split so that validation text is a block the model never sees. Training examples are random windows of 64 characters, with the target at each position being the next character.

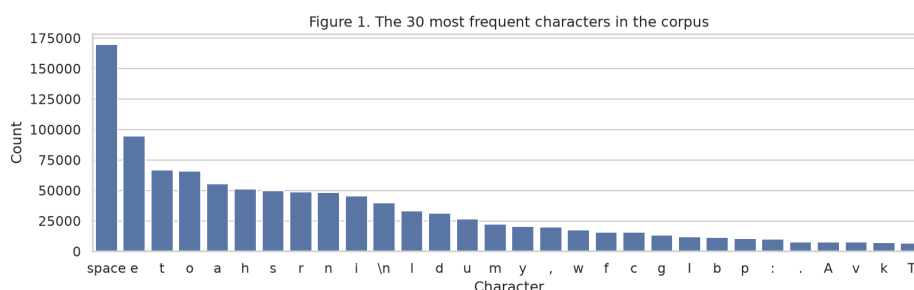


Figure 1. The 30 most frequent characters in the corpus.

3. Model Design and Training Approach

Architecture. The model has 818,241 parameters. A token embedding and a learned position embedding are summed and passed through four identical blocks. Each block contains causal multi-head self-attention with four heads, followed by a two-layer feed-forward network, with layer normalisation applied before each sub-layer and residual connections around both (Ba et al., 2016; Vaswani et al., 2017). A final linear layer produces 65 scores, one for each character. The causal mask is essential: it stops a position from attending to later characters, which would let the model read the answer rather than predict it. I used a small model with a 64-character context because training had to finish in minutes on two CPU cores, and the consequences of that choice are visible in the outputs, as Section 4 shows.

Training. The loss is cross-entropy on the next character, and the optimiser is AdamW, which separates weight decay from the gradient update (Loshchilov & Hutter, 2019). Dropout of 0.1 reduces over-fitting on a corpus of this size (Srivastava et al., 2014). The learning rate warms up linearly for 100 steps and then decays along a cosine curve from 0.003 to 0.0003, and gradients are clipped at a norm of 1.0, which guards against unstable early updates. Training ran for 2,000 steps of 32 windows, about four passes over the data. Training was stable: both losses fell from about 4.3, close to the 4.17 expected from guessing among 65 characters, to a training loss of 1.473 and a validation loss of 1.659 (a perplexity of 5.25). The gap between the two widened from 0.12 at step 400 to 0.19 at the end, a mild sign of over-fitting, and the validation loss was still falling.

Figure 2. Training diagnostics

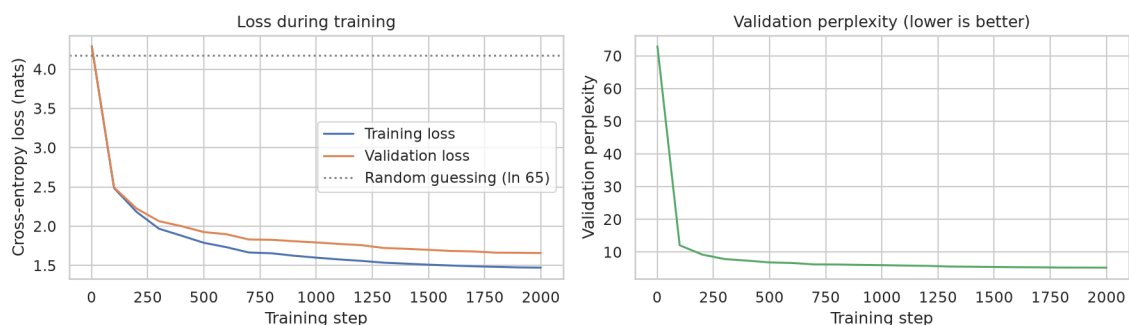


Figure 2. Training and validation loss, and validation perplexity.

4. Output Evaluation and Interpretation

There is no single accepted score for generated text, so I combined three simple measurable properties with a qualitative reading. Sampling temperature, which divides the model's scores before they are turned into probabilities, controls the trade-off between safe, repetitive text and varied, error-prone text (Holtzman et al., 2020), so I evaluated four values on eight 600-character samples each from the prompt 'ROMEO:'. The valid-word rate (the share of generated words found in the training text) fell from 0.953 at temperature 0.5 to 0.670 at 1.3. The distinct-word ratio, a repetition check, rose from 0.554 to 0.872. Temperature 0.8 gave 0.878 and 0.716 and was the best compromise. The figures come from one trained model and one prompt, so they show trends and not precise values.

Temperature	Valid-word rate	Distinct-word ratio	Mean longest verbatim copy (characters)
0.5	0.953	0.554	20.5
0.8	0.878	0.716	19.0
1.0	0.831	0.775	18.6

1.3	0.670	0.872	15.4
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Table 1. Properties of generated text by sampling temperature.

Figure 3. Output properties by sampling temperature

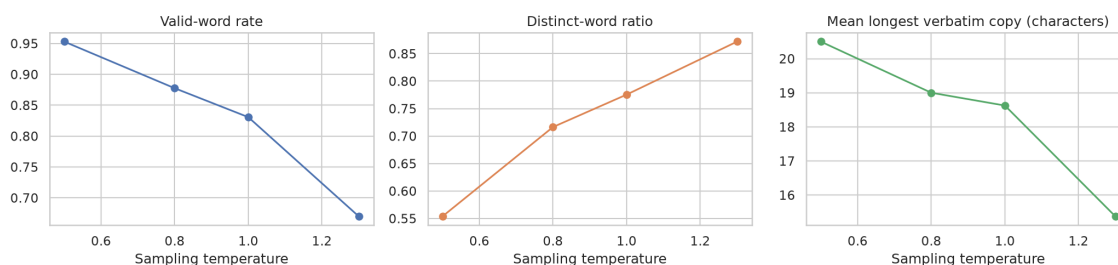


Figure 3. Output properties by sampling temperature.

Strengths. The model learned the layout of a play script. Across eight samples at each temperature, 30 to 34 lines were well-formed speaker labels, and the labels used real names from the corpus, such as MENENIUS, COMINIUS and GLOUCESTER. It also reproduced period spelling and contractions ('Tis, 'twould, hath, doth). **Failure cases.** The text has no coherent meaning. At temperature 0.5 it wrote 'And son the is sons of my black', and at 0.8 it produced the invented words 'worther', 'gived' and 'comforust'. At 1.3 it produced 'franettenm'd' and 'you'ardexile', and a 400-character sample degraded into 'Gitlew' out a careat at eince!'. The model also invented speaker labels that do not exist in the corpus, such as 'Secives:', 'ESANE:' and 'PAURINCE:'. After the prompt 'ROMEO:' it moved at once to speakers from other plays, because its 64-character context cannot hold a scene together. A further behaviour was an artefact of my own setup: three different prompts ('First Citizen:', 'KING HENRY:' and 'DENNIS:') led into almost the same continuation because I used one random seed for all three, which shows that a single fixed-seed sample can mislead. The model accepted the unseen speaker 'DENNIS:' and replied in a normal format, so it has learned the layout as a template and has no knowledge of who the characters are.

5. Ethical Considerations and Responsible Use

Bias in the training data. The corpus is strongly male-weighted: male pronouns outnumber female pronouns by 3.24 to 1, and only two women, Isabella and Juliet, appear among the ten most frequent speakers. A model trained on this text learns the proportions it is shown, and Bender et al. (2021) argue that large text corpora encode the viewpoints of those who produced them and should be documented and examined accordingly. Here the effect is limited because the model is small and the outputs are not meaningful, but the same corpus used to train a larger system for writing or education would reproduce a skewed picture of who speaks and who is spoken about. The corpus also reflects sixteenth and seventeenth century attitudes, so outputs should not be read as neutral.

Misuse and attribution. The generated text imitates the form of a Shakespeare script closely enough that a fragment with a plausible name, such as 'MENENIUS:', could be mistaken for a lost or genuine passage, while the invented labels ('ESANE:', 'PAURINCE:') and invented words show that the content is fabricated. This is a small instance of hallucination, where a generator produces fluent but false material. A plausible misuse is to present generated verse as authentic or as a student's own writing. Responsible use therefore means labelling outputs as machine-generated and never presenting them as original text of the plays.

Creative ownership and memorisation. The plays are in the public domain, so training on them raises no copyright problem here. The same technique applied to copyrighted work would, because language models can memorise and reproduce training text (Carlini et al., 2021). I tested for this by measuring the longest string copied unchanged from the training text: it averaged 15 to 21 characters and never exceeded 29, and the longest span was a stock sequence of dialogue formatting, not a line of verse. The prompt 'To be, or not to be' was not continued with the famous line. Within this small test I found no reproduction of passages, but a small model trained for four passes over the data says little about how a larger model would behave, so the same test would need repeating before applying the approach to protected material.

6. Limitations and Future Improvements

Limitations. The model is small (818,241 parameters) and sees only 64 characters at a time, which explains the missing coherence and the drift away from the prompted speaker. Training stopped at 2,000 steps while the validation loss was still falling. The evaluation has limits of its own: the measures were taken on eight samples per temperature from one prompt and one trained model, the valid-word rate cannot judge meaning, and I did not carry out a human rating of quality. Valid-word counting also treats any word found in the training text as correct, even when it is used wrongly. The edition of the corpus could not be verified, and the corpus is male-weighted and in archaic English.

Future improvements. The first step is a longer context and more training steps, so that the model can keep a speaker and a scene consistent. Training on a larger and more balanced set of dramatic texts, with the edition documented, would address the imbalance I measured. For evaluation, I would add human ratings of fluency and coherence by several readers, and a comparison with nucleus sampling (Holtzman et al., 2020), which is designed to keep variety without the loss of word validity seen at high temperature. Repeating the memorisation test at larger model sizes would show whether the low copying found here holds as capacity grows. Every generated sample should also carry a machine-generated label in any real use.

References

- Ba, J. L., Kiros, J. R., & Hinton, G. E. (2016). Layer normalization. *arXiv*. <https://arxiv.org/abs/1607.06450>
- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610-623. <https://doi.org/10.1145/3442188.3445922>
- Carlini, N., Tramer, F., Wallace, E., Jagielski, M., Herbert-Voss, A., Lee, K., Roberts, A., Brown, T., Song, D., Erlingsson, U., Oprea, A., & Raffel, C. (2021). Extracting training data from large language models. *Proceedings of the 30th USENIX Security Symposium*, 2633-2650. <https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting>
- Holtzman, A., Buys, J., Du, L., Forbes, M., & Choi, Y. (2020). The curious case of neural text degeneration. *8th International Conference on Learning Representations (ICLR 2020)*. <https://arxiv.org/abs/1904.09751>
- Karpathy, A. (2015). *char-rnn: Multi-layer recurrent neural networks (LSTM, GRU, RNN) for character-level language models in Torch* [Computer software and data set]. GitHub. <https://github.com/karpathy/char-rnn>
- Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *7th International Conference on Learning Representations (ICLR 2019)*. <https://arxiv.org/abs/1711.05101>

- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Kopf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., & Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32. <https://arxiv.org/abs/1912.01703>
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). *Language models are unsupervised multitask learners* [Technical report]. OpenAI. https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56), 1929-1958. <https://jmlr.org/papers/v15/srivastava14a.html>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *arXiv*. <https://arxiv.org/abs/1706.03762>